Simulation Arena

Approval Bottleneck: Where Deals Actually Get Stuck

CF Approval is the jam: 87.4% busy, adding 3.4 days of waiting, and it is why 50.7% of requests miss the 10-day promise. $3.02M of revenue is stuck behind it. Auto-approve requests under 10% would release $2.65M of that.

Seed 42 | Run run_1f5b9a795516 | 4,000 scenarios tested | Discrete-event queueing simulation (arr… | 21 Sep 2026, 21:34

Same seed, same numbers - this run can be reproduced exactly.

How this works

We rebuilt your approval process as a queue. Requests arrive the way yours do, each waits for a free approver, gets worked on for a realistic amount of time, then moves to the next step - and a request that waits too long sometimes gives up and walks. Running thousands of requests through that queue shows where the pile-up forms, how long the whole journey takes, and how many requests would still be sitting there when the quarter closes. We then re-ran the same queue with each fix in place to price them against each other. Every option is run on the same dice - same arrivals, same requests - so the difference between two options is the fix itself rather than luck.

The numbers that matter

HOW LONG A REQUEST TAKES TODAY
10.1 days
Half of all requests clear in 10.1 days or less.
vs promise is 10 days

CLEARED INSIDE THE 10-DAY PROMISE
49.3%
The share of requests that still land in the quarter they were meant to.
vs 50.7% miss it

WHERE IT JAMS
CF Approval
This step is 87.4% busy and adds 3.4 days of waiting - the single biggest cause of the delay.

BUSIEST STEP IS BUSY
87.4%
Above about 85% a queue stops clearing: any hiccup becomes a backlog.
vs 3.4 days of waiting

REVENUE STUCK BEHIND THE QUEUE
$3.02M
Deal value of the requests that miss the quarter, times the 15.0% you told us is lost when a deal slips.
vs $38.36M of deals pass through in a quarter

WALK-AWAY RATE
0.0%
Requests that waited more than 25 days and gave up.

BEST FIX: AUTO-APPROVE REQUESTS UNDER 10%
-3.5 days
Median time goes from 10.1 to 6.6 days and requests inside the promise go from 49.3% to 94.0%.
vs $2.65M of revenue released

How we got there

1. The process we rebuilt

10.2 requests a week arrive and pass through Deal Desk Review -> CF Approval -> Legal, with 6, 4, 5 approvers respectively. Measured from 260 requests in your own history, spanning 179 days.

Evidence: $38.36M of deals through the queue per quarter

2. Where it jams

CF Approval is 87.4% busy and makes requests wait 3.4 days on average - 33.2% of the whole journey is spent sitting in front of that one step.

Evidence: 87.4% busy, 3.4 days waiting

3. What that costs

Only 49.3% of requests clear inside the 10-day promise, so 50.7% slip out of the quarter. At 15.0% of deal value lost per slip, that is $3.02M of revenue at risk. Nothing gives up outright, but the slowest tenth wait 17 days.

Evidence: $3.02M at risk

4. What the fixes are worth

Auto-approve requests under 10% is the best of them: median time falls from 10.1 to 6.6 days, requests inside the promise rise from 49.3% to 94.0%, and $2.65M of revenue comes back into the quarter.

Evidence: $2.65M released

5. The bottom line

The constraint is not effort, it is capacity at CF Approval: the work itself takes 2.7 days, the waiting takes 3.4. Add capacity or take work out of that queue - asking people to work faster will not move the number.

Evidence: 2.7 days work, 3.4 days waiting

Charts

How long requests actually take

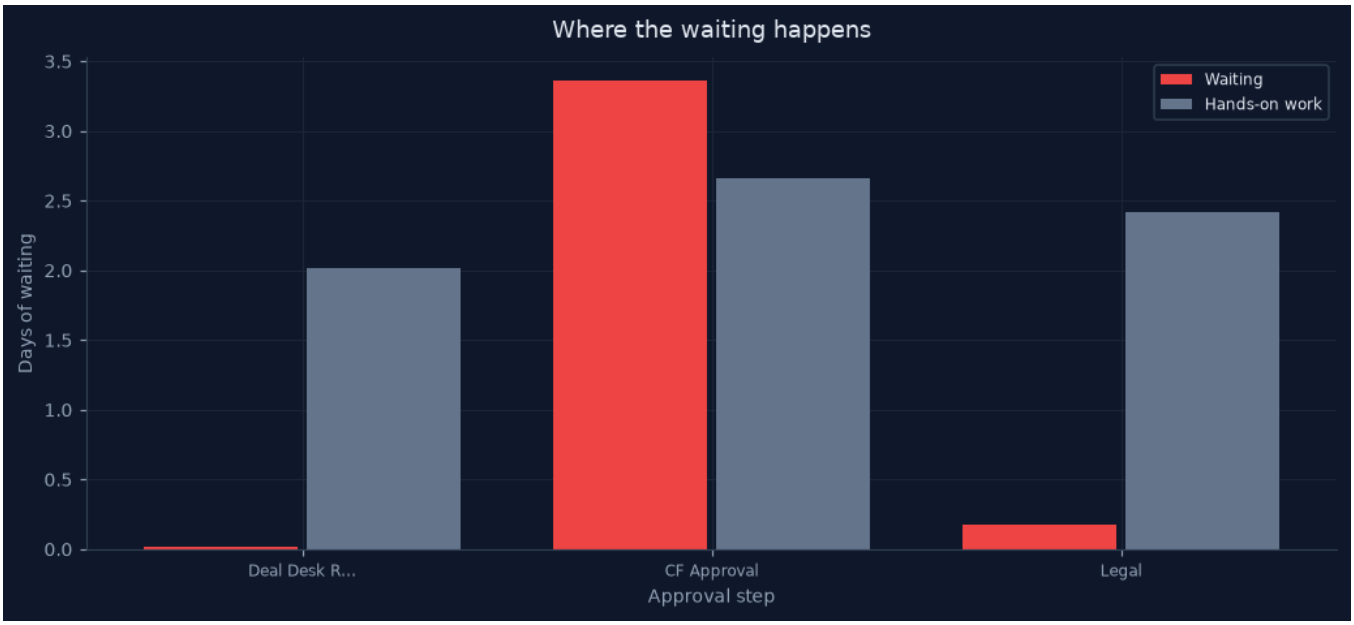
Today's process; the promise is 10 days



What this shows: Everything to the right of the promise line is a deal that slips out of the quarter. The long tail on the right is the queue, not the work.

Where the waiting happens

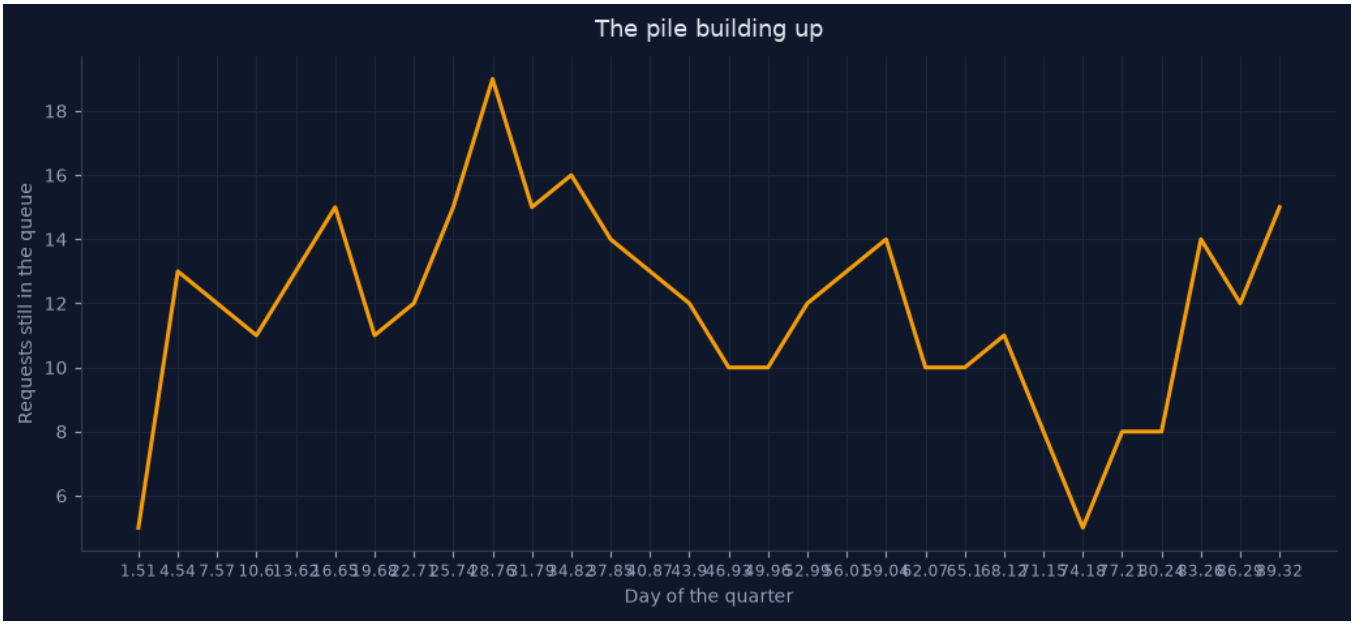
Average days a request sits before each step even starts



What this shows: The tall red bar is the bottleneck. Grey is the actual work - if red dwarfs grey, the problem is capacity, not effort.

The pile building up

Requests open at any point in the quarter



What this shows: A line that keeps climbing means the queue never catches up - that is the definition of a bottleneck.

Tables

Each step, measured

Waiting is the queue; hands-on is the actual work. 'Days in your file' is the door-to-door time you recorded - waiting and work together - which is why it is bigger than hands-on.

Step	Approvers	Requests	Days in your file	Avg wait (days)	Slowest 10% wait	Longest wait	Hands-on (days)	How busy	Share of all waiting	Column 11
Deal Desk Review	6	649	2.1	0.0	0.0	0.9	2.0	47.6%	0.6%	-
CF Approval	4	649	7.6	3.4	6.5	15.3	2.7	87.4%	94.2%	BOTTLENECK
Legal	5	649	2.7	0.2	0.7	2.5	2.4	60.7%	5.2%	-

The bottleneck step is the one with the most waiting per unit of busyness - it is the reason the quarter slips.

What each fix is worth

Revenue at risk = deal value of the requests that miss the 10-day promise, times 15.0%.

Option	Days to clear (median)	Slowest 10% (days)	Inside the promise	Walk-away rate	Busiest step busy	Revenue at risk	Gain
Today's process	10.1	16.7	49.3%	0.0%	87.4%	$3.02M	-
One more approver in CF Approval	8.1	11.3	77.2%	0.0%	73.1%	$1.40M	$1.62M released
Auto-approve requests under 10%	6.6	9.3	94.0%	0.0%	64.1%	$366k	$2.65M released
One more approver in Legal	9.8	16.5	51.9%	0.0%	87.4%	$2.85M	$168k released

The highlighted row is the cheapest fix that releases the most revenue.

The ten worst-hit requests

The requests that lost the most time in today's process.

Request	Arrived (day)	Days to clear	Of which waiting	Stuck behind	Deal value	Value at risk	Outcome
Request 64	39	24.1	8.1	CF Approval	$349k	$52k	Late
Request 104	61	22.1	13.5	CF Approval	$743k	$111k	Late
Request 135	86	21.5	10.3	CF Approval	$230k	$34k	Late
Request 103	60	21.4	15.4	CF Approval	$391k	$59k	Late
Request 100	59	21.4	14.5	CF Approval	$717k	$108k	Late
Request 133	83	21.3	11.7	CF Approval	$788k	$118k	Late
Request 105	61	20.9	14.8	CF Approval	$171k	$26k	Late
Request 81	49	20.6	10.3	CF Approval	$235k	$35k	Late
Request 96	58	20.5	11.4	CF Approval	$81k	$12k	Late
Request 97	58	20.5	12.1	CF Approval	$209k	$31k	Late

Multiply this pattern across the quarter and you have the size of the problem.

What we recommend

Fix CF Approval first. Auto-approve requests under 10% releases $2.65M of revenue and takes 3.5 days out of the process.

Risk Moderate 51 out of 100 | Confidence: High

Why

- CF Approval runs at 87.4% busy and adds 3.4 days of waiting - more than any other step.
- Only 49.3% of requests clear inside the 10-day promise, putting $3.02M of revenue at risk.
- The work in that step takes 2.7 days; the waiting takes 3.4. It is a capacity problem, not an effort problem.

What to do

1. Auto-approve requests under 10% - worth $2.65M of revenue released and 3.5 days off the median.

2. Auto-approve anything under 10% discount without a CF Approval stop - that alone removes 40.0% of the queue.

3. Publish a daily view of the CF Approval queue with a 10-day clock on each request, so slippage is visible before the quarter closes.

4. Watch the tail: the slowest requests take 17 days. Nothing is walking away yet, but at that length the customer has already re-planned without you.

The trade-off: Adding an approver costs salary; auto-approving below a threshold costs a little margin on small deals. Both are far cheaper than the $3.02M currently at risk.

What we assumed and what to watch

Worth knowing before you act

• Fewer than half of requests clear inside the 10-day promise. Until the bottleneck is fixed, the forecast for this quarter is optimistic.

Assumptions

• Arrivals and timing measured from your own approval history: about 10.2 requests a week, 3 steps.

• Hands-on time per step: Deal Desk Review about 2.1 days; CF Approval about 2.7 days; Legal about 2.4 days

• Where your file records door-to-door time (queue included), we backed out the real hands-on time by finding the working time that reproduces the waiting you actually saw, given how many approvers you have.

• Approvers per step: Deal Desk Review 6, CF Approval 4, Legal 5 (counted from the distinct names in your file)

• A request that waits more than 25 days gives up with 25.0% probability.

• Revenue at risk = 15.0% of the deal value of every request that misses the 10-day promise; average deal value $288,286.

• Every request is modelled as passing through every step; steps that only some requests reach therefore look slightly busier than they are.

• 5 simulated quarters of 200 requests each per option (1,000 requests in total); about 129 arrive and clear inside each 90-day quarter. Seed 42, so re-running gives identical numbers. Every option is tested against the same simulated requests, so the comparison is fair. Revenue at risk is the average per quarter.

• Request / ticket ID -> your 'request_id' column

• Day it was submitted -> your 'submitted_day' column

• Approval step -> your 'stage' column

• Days it spent in that step -> your 'days_in_stage' column

• Who approved it -> your 'approver' column

• Deal value -> your 'deal_value' column

Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

Time	Stage	What happened
0.00s	prepare	You chose 'stage' for 'Approval step'.
0.00s	prepare	You chose 'days_in_stage' for 'Days it spent in that step'.
0.00s	prepare	You chose 'approver' for 'Who approved it'.
0.00s	prepare	You chose 'deal_value' for 'Deal value'.
0.00s	prepare	Request / ticket ID -> your 'request_id' column
0.00s	prepare	Day it was submitted -> your 'submitted_day' column
0.00s	prepare	Approval step -> your 'stage' column
0.00s	prepare	Days it spent in that step -> your 'days_in_stage' column
0.00s	prepare	Who approved it -> your 'approver' column
0.00s	prepare	Deal value -> your 'deal_value' column
0.02s	prepare	Read 709 records into the model.
0.02s	simulate	Running the simulation…
0.02s	read	Reading how approval requests arrive and how long each step takes...

Time	Stage	What happened
0.14s	prepare	'Deal Desk Review': 260 requests, 6 approver(s), 2.1 days door to door - about 2.1 of that is real work.
0.27s	prepare	'CF Approval': 260 requests, 4 approver(s), 7.6 days door to door - about 2.7 of that is real work.
0.39s	prepare	'Legal': 189 requests, 5 approver(s), 2.7 days door to door - about 2.4 of that is real work.
0.39s	prepare	Learned the process from 709 request rows covering 179 days - about 1.45 requests a day.
0.39s	prepare	Rebuilding the queue: 10.2 requests a week, 6 in Deal Desk Review, 4 in CF Approval, 5 in Legal.
0.39s	simulate	Running 1,000 requests through the queue (5 simulated quarters)...
0.40s	summarize	Found it: CF Approval is the jam - 87.4% busy and 3.4 days of waiting.
0.42s	simulate	Tested 'One more approver in CF Approval'.
0.44s	simulate	Tested 'Auto-approve requests under 10%'.
0.47s	simulate	Tested 'One more approver in Legal'.
0.47s	summarize	Comparing the fixes...
0.47s	summarize	Done. CF Approval is the bottleneck - $3.02M of revenue is stuck behind it.